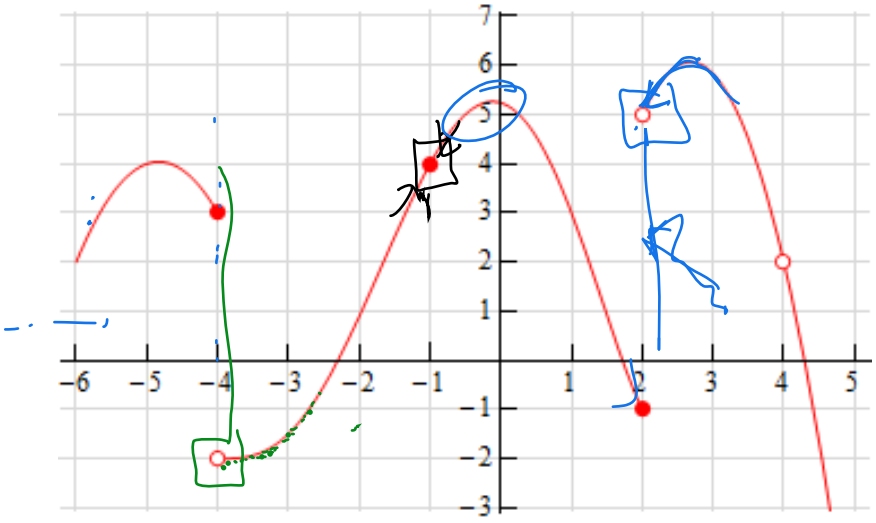


○ $x \neq 2$ $(2, \ldots$

● $x = 2$ $[2,$

one-sided
$\lim_{x \to -4^-} f(x) = 3$ meaning as $x \to -4^-$, $y \to 3$

$\lim_{x \to -4^+} f(x) = -2$ meaning as $x \to -4^+$, $y \to -2$

two-sided: $\lim_{x \to -4} f(x) =$ does not exist because $\lim_{x \to -4^-} f(x) \neq \lim_{x \to -4^+} f(x)$

$f(-4) = 3$ (functional value)

two-sided $\lim_{x \to -1} f(x) = 4$

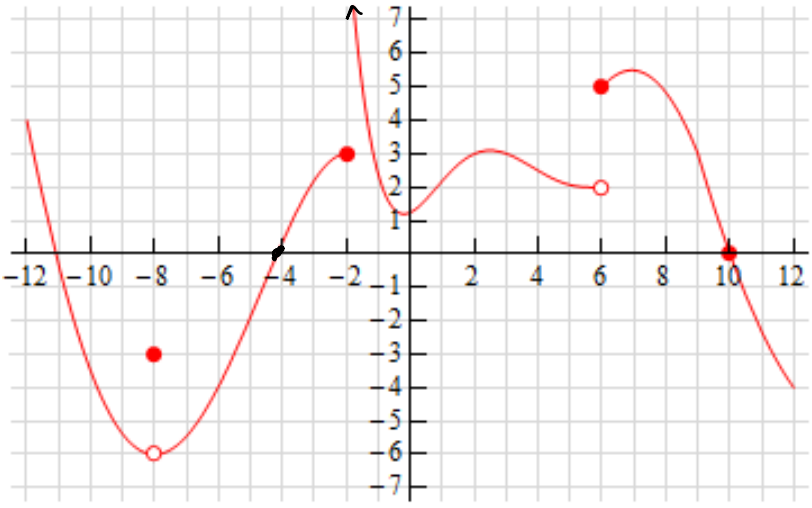
$f(-1) = 4$ (functional value)

one-sided
$\lim_{x \to 2^-} f(x) = -1$

$\lim_{x \to 2^+} f(x) = 5$

two-sided $\lim_{x \to 2} f(x) =$ do is not exist because $\lim_{x \to 2^-} f(x) \neq \lim_{x \to 2^+} f(x)$

$f(2) = -1$ (functional value)



$\lim_{x \to -8} f(x) = -6$

$f(-8) = -3$

$\lim_{x \to -4} f(x) = 0$

$f(-4) = 0$

$\lim_{x \to -2^-} f(x) = 3$

$\lim_{x \to -2^+} f(x) = +\infty$ (does not exist)

$\lim_{x \to -2} f(x) =$ does not exist because ① $\lim_{x \to -2^+} f(x)$ does not exist ② $\lim_{x \to -2^-} f(x) \neq \lim_{x \to -2^+} f(x)$

$f(-2) = 3$

$\lim_{x \to 6^-} f(x) = 2$

$\lim_{x \to 6^+} f(x) = 5$

$\lim_{x \to 6} f(x) =$ does not exist because $\lim_{x \to 6^-} f(x) \neq \lim_{x \to 6^+} f(x)$

$f(6) = 5$